

Multiple Choice (10 marks)

1. Which of the following are the spatial clustering algorithms?
 - (a) Partitioning based clustering
 - (b) K-means clustering
 - (c) Grid based clustering
 - (d) All of the above
2. Which of the following is NOT supervised learning?
 - (a) PCA
 - (b) Decision Tree
 - (c) Linear Regression
 - (d) Naive Bayesian
3. Statement 1| Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2| DenseNets usually cost more memory than ResNets.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 3
 - (b) 4
 - (c) 7
 - (d) 15
5. Which of the following statements about Naive Bayes is incorrect?
 - (a) Attributes are equally important.
 - (b) Attributes are statistically dependent of one another given the class value.
 - (c) Attributes are statistically independent of one another given the class value.
 - (d) Attributes can be nominal or numeric

True / False (10 marks)

Answer True or False

6. True or False: Bayesians advocate for the use of prior distributions on parameters in probabilistic models, while frequentists do not.
7. True or False: In Yann LeCun's framework, self-supervised learning is regarded as the most critical component in machine learning.
8. True or False: Residual connections are utilized in both ResNets and Transformers to enhance model performance.
9. True or False: The variance of the Maximum A Posteriori (MAP) estimate is lower than that of the Maximum Likelihood Estimate (MLE).
10. True or False: The ResNet-50 model contains over 1 billion parameters, making it one of the largest neural networks available.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the maximum total path sum in the given triangle.
12. Write a function that matches a string that has an a followed by three 'b'.

End of Paper